

Analyst Intern Project Brief



Outline

Objective: predict NBA shot outcomes using features derived from player tracking data.

Provided is a sample of shot attempts in NBA regular season games for a few recent seasons. Your task is to:

- use the training data (**training.csv.gz**) to create a model that predicts shot make probabilities (between 0 and 1) for all provided shots in the testing dataset (**testing.csv.gz**)
- write up a short summary of how your model works. Please include a printout, visualization, and/or short explanation that explains the features it uses and prioritizes (e.g. regression model summary, variable importance plot, description of model architecture/features etc.). You are encouraged to add reflections on what you attempted to do, any pitfalls you ran into, and what additional features you think would improve your model.

All data required to complete the project are contained in this GitHub repository. You are not allowed to use any external sources of data. Please use R (preferred) or Python as the primary programming language to complete this project. Your work must be your own and original. You may use AI tools to aid your work if you include a log with your prompts (more details provided below).

Assessment

Your predicted probabilities on shots in **testing.csv.gz** will be evaluated against probabilities outputted by our in-house shot make model, which was trained using many of the same features. Error will be computed on log-loss.

Of the top scoring participants, we will invite a select group to a technical interview as the next part of our internship hiring process. Our evaluation of who to select for interviews will incorporate both the quality of predictions and the quality of thought indicated in the write-up. Among similar scoring responses, we will prioritize those with more explainability and better process in their write ups.

You should spend no more than 20 hours on this project.

Submission Instructions

Your submission will be evaluated holistically, incorporating the accuracy of your predictions on the testing dataset, the quality of your writeup explaining your model, and evidence of well-organized, readable code.

Your GitHub repo must contain the following four files at the top level (not in a subfolder), named exactly as shown:

1. **submission.csv**: your predictions
 - A template is already in your repository. Fill in the **make_prob** column with your model's shot predictions.
 - Do not add, remove, or rename any columns.
 - Do not reorder any of the rows. **shot_id** is already in the same order as the corresponding column in **testing.csv.gz**.
2. **project_code**. (R|Rmd|py|ipynb): your code
 - A single file containing everything needed to fit your model and fill in predictions in **submission.csv**.
 - It is okay to include a small number of helper files, but the main script must be titled **project_code**.
 - Please put your name and the last 4 digits of the phone number used on your application in a comment at the top of **project_code** in the form **#NAME: (your name here) | NUMBER: (your last 4 digits)**.
3. **project_writeup**. (PDF|html): your project writeup
 - A single file containing your written description of your model, as well as any accompanying analysis / discussion. A PDF is preferred, but an HTML is acceptable too.
 - There is no required format or minimum length, but we are looking for quality discussion and visuals to explain the structure of your model, the features you decided to use, your outputs, and thoughts you have on how you may improve your model with more time and/or more data.
4. **ai_prompts**. (md|txt): your AI prompts
 - If you leveraged AI tools to help complete the project, a single file containing all prompts you used.

Note: please do not upload any files into the repo exceeding 10 MB. The only files permitted above 10 MB are the train and test data already in the repo. You may choose to upload image files so that they render in your writeup, but please mind the 10MB limit.

Once you are done with the project, commit all changes to the main branch, then hit "Submit project" in the email you received to start the project process. We will not be evaluating any files edited after you submit your project.

Please consult the following checklist prior to clicking "Submit project":

- All four files are at the top level of your repo.
- **submission.csv** has **make_prob** filled in for every row, with the original columns and row order intact.
- Your writeup renders correctly with all intended images.

Submissions that fail to adhere to these guidelines will not be reviewed.

Dataset information

training.csv (compressed to a gzip: please do not unzip)

a dataset of 425,719 non-fouled regular season shots with the following features:

- **shot_id**: string identifier unique to the shot
- **outcome**: binary, whether the shot was made (1) or not (0)
- **season_id**: string identifier representing the season
- **off_team_id**: string identifier representing the offensive team
- **def_team_id**: string identifier representing the opposing, defensive team
- **shooter_id**: string identifier representing the shooter
- **closestdef_id**: string identifier representing the closest defender (note: the closest defender may or may not be labeled as a contesters of the shot)
- **num_contesters**: integer, the number of contesters of the shot
- **contester1_id, ..., contester4_id**: string identifiers representing the contesters of the shot
- **distcont1, ..., distcont4**: float, distances (in feet) of the contesters of the shot
- **contested**: boolean, whether the shot was contested
- **month**: string, the month of the game
- **gamestate**: string representing the game context of the shot (one of halfcourt, transition, or direct second chance)
- **shottype**: string representing the type of shot (one of dunk, layup, tip, jumper, layup, floater, post, heave)
- **distance**: float, distance (in feet) of the shot from the basket
- **locationx, locationy**: float coordinates of the shot in feet (x: along the sideline, y: along the baseline, see court diagram below)
- **three**: boolean, whether the shot was a 3 point attempt or not
- **dribblesbefore**: integer, the number of dribbles by the shooter prior to the shot
- **shotclock**: float, seconds remaining on the shot clock
- **closestdefdist**: float, distance of closest defender to shooter at the time of release
- **closestdefapproach**: array of floats representing the closest defender's distance at 1 second, 0.75 seconds, 0.5 seconds, and 0.25 seconds before the shot
- **shooterspeed**: float, speed of the shooter at the time of release (feet per second)

testing.csv (compressed to a gzip: please do not unzip)

a dataset of 213,977 non-fouled regular season shots with the same features as **training.csv**, but without the **outcome** column

submission.csv

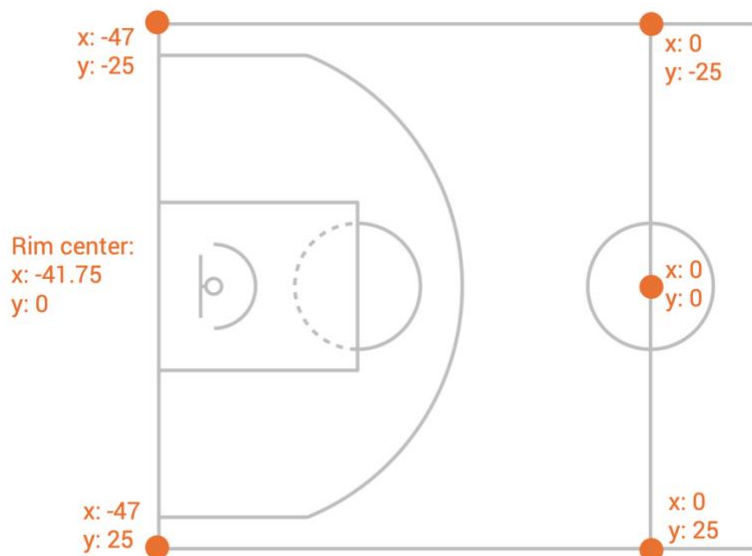
a template submission file with two columns:

- **shot_id**
- **make_prob**

Your submission **must** be formatted the same way with the same column names.

Reference

The court diagram below can help anchor **locationx** (the coordinate along the sideline) and **locationy** (along the baseline):



See the commands below for examples of reading in compressed data. **Please do not unzip the training/testing .csv.gz files.**

Using R:

```
library(readr)
train_df <- read_csv("data/training.csv.gz")
```

Using Python:

```
import polars as pl
train_df = pl.read_csv("data/training.csv.gz")
```